

Part A – Graph Creation and Modification (30 pts)

Q1: Graph Creation (15 pts)

The graph shown in the assignment is an undirected graph on the ten vertices $V = \{0, 1, \dots, 9\}$. Reading the edges off the figure gives $|E| = 15$ edges:

$$E = \left\{ \begin{array}{l} (0, 1), (0, 4), (0, 5), (1, 2), (1, 6), \\ (2, 3), (2, 7), (3, 4), (3, 8), (4, 9), \\ (5, 7), (5, 8), (6, 8), (6, 9), (7, 9). \end{array} \right\}$$

Equivalently, as an adjacency list:

Node	Neighbours	Node	Neighbours
0	1, 4, 5	5	0, 7, 8
1	0, 2, 6	6	1, 8, 9
2	1, 3, 7	7	2, 5, 9
3	2, 4, 8	8	3, 5, 6
4	0, 3, 9	9	4, 6, 7

Construction. The graph was built in NetworkX by creating an empty undirected graph, adding the ten nodes $0, \dots, 9$, and then adding the fifteen edges listed above:

```
G = nx.Graph()
G.add_nodes_from(range(10))
G.add_edges_from(E)
```

Checks on the transcription. Every vertex has degree exactly 3, so the graph is 3-regular and the handshake lemma is satisfied: $\sum_v \deg(v) = 30 = 2|E| = 2 \cdot 15$. The graph is connected, has no triangles, and has girth 5. In fact these fifteen edges reproduce the *Petersen graph* (checked with NetworkX's `is_isomorphic` against `petersen_graph()`, which returns `True`), the standard 3-regular, girth-5 graph on ten vertices. Landing on the Petersen graph is a strong confirmation that the edge list was read off the drawing correctly.

One edge deserves comment, because the drawing in the assignment is misleading at that point: vertices 5, 4 and 8 are very nearly collinear, and the single edge $(5, 8)$ is drawn passing straight across the disc of vertex 4. It therefore *looks* like a pair of edges $(5, 4)$ and $(4, 8)$, but it is one edge; if it were two, vertices 5, 4 and 8 would all have degree 4 and the graph would

not be 3-regular. In Figure 1 the edge $(5, 8)$ is drawn with a slight bow so that it visibly passes clear of vertex 4.

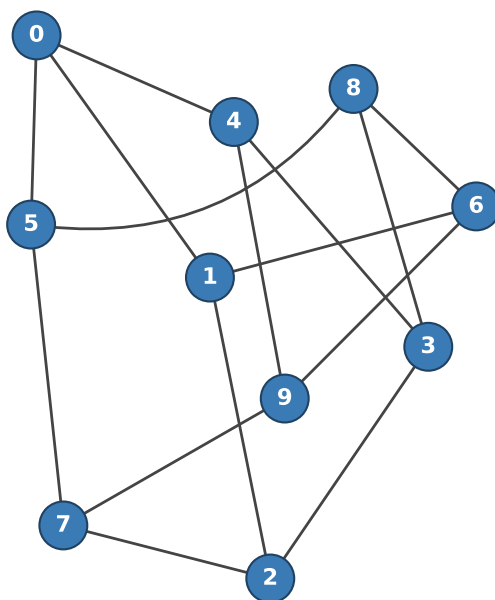


Figure 1: Q1: the undirected graph on 10 vertices and 15 edges, drawn with the same vertex placement as the figure in the assignment. The edge $(5, 8)$ is bowed so that it is visibly distinct from the vertex 4 it passes behind.

The visualization is shown in Figure 1. It uses the same vertex positions as the drawing in the assignment, so the two can be compared directly.

Q2: Graph Modification (15 pts)

Starting from the graph G of Q1 we remove, in order,

1. **node** 1, which also deletes its three incident edges $(0, 1)$, $(1, 2)$ and $(1, 6)$;
2. **edge** $(0, 4)$;
3. **edge** $(0, 5)$.

Five edges are removed in total, leaving the graph H on nine vertices with ten edges:

$$E(H) = \{(2, 3), (2, 7), (3, 4), (3, 8), (4, 9), (5, 7), (5, 8), (6, 8), (6, 9), (7, 9)\}.$$

	Before (Q1)	After (Q2)
Nodes	10	9
Edges	15	10
Connected components	1	2

Remaining components. The three edges at node 0 were exactly $(0, 1)$, $(0, 4)$ and $(0, 5)$, and all three are removed by the three operations above. Node 0 is therefore left isolated, and the resulting graph has **two** connected components:

Component	Vertices	Size
C_1	$\{0\}$	1
C_2	$\{2, 3, 4, 5, 6, 7, 8, 9\}$	8

C_2 contains all ten remaining edges and is connected; C_1 is the isolated vertex 0. The remaining graph is drawn in Figure 2, using the same vertex positions as Figure 1 so that the effect of the modification is directly visible; the isolated vertex 0 is drawn in orange.

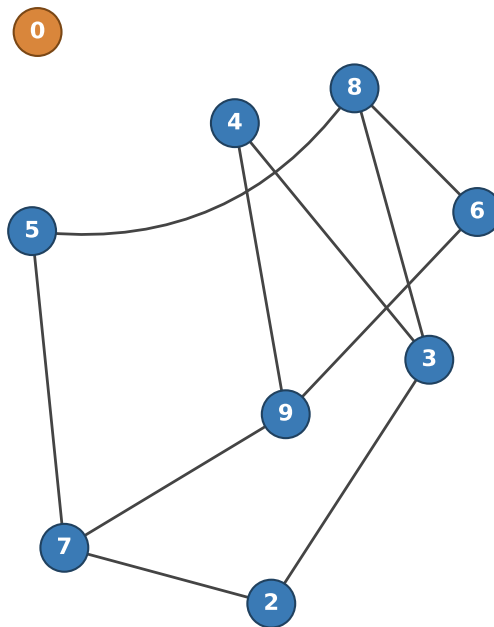


Figure 2: Q2: the graph after removing node 1 and the edges $(0,4)$ and $(0,5)$. Vertex positions are unchanged from Figure 1. The two connected components are the isolated vertex 0 (orange) and the remaining eight vertices.